### 23.1.1 Exercises - Mass defect

1. Study the decay equations given in Worked examples 1 and 2, and write balanced equations for the following:

   a. A nucleus of radon-220 ($^{220}_{86}Rn$) decays by α emission to form an isotope of polonium, Po.

   b. A nucleus of a sodium isotope ($^{25}_{11}Na$) decays by β− emission to form an isotope of magnesium, Mg

   c. Copy and complete this equation for the β− decay of a nucleus of argon:

$$^{41}_{18}Ar \rightarrow K + ?$$

2. The Sun releases vast amounts of energy. Its power output is $4.0 \times 10^{26}$ W. Estimate how much its mass decreases each second because of this energy loss.

$\Delta m$ = ..... kg

3. The masses of the particles are given in Table.

| Particle | Mass / $10^{-27}$ kg |
|---|---|
| $^{1}_{1}p$ | 1.672 623 |
| $^{1}_{0}n$ | 1.674 929 |
| $^{4}_{2}He$ | 6.644 661 |

Calculate the energy released per nucleon if a $^{4}_{2}He$ nucleus is formed from separate stationary protons and neutrons.

energy release per nucleon= ..... J

4. The rest mass of a golf ball is 150 g. Calculate its increase in mass when it is travelling at 50 m $s^{-1}$. What is this as a percentage of its rest mass?

percentage of its rest mass = ..... %

5.

a. The mass of an atom of $^{56}_{26}Fe$ is 55.934 937 u. Calculate its mass in kg.

mass = ..... kg

b. The mass of an atom of $^{16}_{8}O$ is 2.656 015 × $10^{-26}$ kg. Calculate its mass in u.

mass = ..... u

6.

a. Explain why the mass of an α-particle is less than the total mass of two individual protons and two individual neutrons.

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b. An equation for one possible nuclear reaction is

$$^{4}_{2}He + ^{14}_{7}N \rightarrow ^{17}_{8}O + ^{1}_{1}p.$$

Data for the masses of the nuclei are given in Fig.

| | | mass/u |
|---|---|---|
| proton | $^{1}_{1}p$ | 1.00728 |
| helium-4 | $^{4}_{2}He$ | 4.00260 |
| nitrogen-14 | $^{14}_{7}N$ | 14.00307 |
| oxygen-17 | $^{17}_{8}O$ | 16.99913 |

i. Calculate the mass change, in u, associated with this reaction.

mass = ..... u [2]

ii. Calculate the energy, in J, associated with the mass change in (i).

energy = ..... J

iii. Suggest and explain why, for this reaction to occur, the helium-4 nucleus must have a minimum speed.

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